Project: Worked examples
Title: Lateral-torsional buckling and N+M interaction (EN 1993-1-1)
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1. Scope and method

Calculates the lateral-torsional buckling, flexural buckling and bending-with-compression interaction of a uniform doubly-symmetric I-section member on fork supports, to EN 1993-1-1. Class 4 sections are outside this sheet.

WHAT IT CALCULATES	RELATION
Critical moment (NCCI SN003 form)	$M_{cr} = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 \cdot I_w/I_z} + (kL)^2 \cdot G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2) - C_2 z_g]$; $C_1 = 1.88 - 1.40\psi + 0.52\psi^2 \le 2.70$ (end moments, k = 1) unless entered
LTB slenderness	$\bar{\lambda}_{LT} = \sqrt{(W_y \cdot f_y/M_{cr})}$, $W_y = W_{pl,y}$ (class 1, 2) or $W_{el,y}$ (class 3)
Lateral-torsional buckling, general case (6.3.2.2)	$\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - 0.2) + \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \bar{\lambda}_{LT}^2)}) \le 1$
Lateral-torsional buckling, rolled sections (6.3.2.3)	$\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \cdot \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \beta \cdot \bar{\lambda}_{LT}^2)}) \le 1$, $\le 1/\bar{\lambda}_{LT}^2$; optional $\chi_{LT,mod} = \chi_{LT}/f \le 1$, $\le 1/\bar{\lambda}_{LT}^2$ (eq. 6.58), $f = 1 - 0.5(1 - k_c)[1 - 2(\bar{\lambda}_{LT} - 0.8)^2] \le 1$
Buckling resistance in bending	$M_{b,Rd} = \chi_{LT} \cdot W_y \cdot f_y/\gamma_{M1}$
Flexural buckling	$N_{cr} = \pi^2 EI/L_{cr}^2$, $\bar{\lambda} = \sqrt{(A \cdot f_y/N_{cr})}$, $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$, $N_{b,Rd} = \chi \cdot A \cdot f_y/\gamma_{M1}$; the curve of each axis from Table 6.2 (section type, h/b, t_f; $f_y \le 420$ MPa) or chosen
Interaction (6.3.3(4))	$N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1}) + k_{yy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{yz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$; $N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1}) + k_{zy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{zz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$
Factors k_{ij}	Annex B from the slenderness, $N_{Ed}/(\chi \cdot N_{Rk}/\gamma_{M1})$ and C_m ($C_m = 0.6 + 0.4\psi \ge 0.4$ for a linear diagram, or entered); Annex A from the μ, w, n_{pl}, a_{LT} and C_{ij} expressions (class 1, 2: plastic column of Table A.1; class 3: elastic column, without the $0.6\sqrt{(w_i/w_j)}$ factors of k_{yz}, k_{zy})

Basis: Buckling curves; slenderness for flexural buckling[1](cl. 6.3.1.2, 6.3.1.3); Lateral-torsional buckling curves: general case[1](cl. 6.3.2.2); Lateral-torsional buckling curves for rolled sections and equivalent welded sections[1](cl. 6.3.2.3); Uniform members in bending and axial compression[1](cl. 6.3.3); Interaction factors k_ij: methods 1 and 2[1](Annex A, Annex B); Elastic critical moment of a doubly-symmetric member[2](Mcr expression); Stability of members (label only, see the edition note)[1](cl. 8.3).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
edition	EN 1993-1-1 edition (the 2022 choice changes the printed label only)	2005+A1:2014		spec: BS EN 1993-1-1:2005+A1:2014 with the UK National Annex	any
ltb_{method}	Method for the lateral-torsional buckling reduction factor	6.3.2.3 (rolled and welded sections)		assumption: 6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd.	any
annex	Annex for the interaction factors	Annex B (method 2)		assumption: Annex B (method 2) chosen; Annex A can be selected for comparison.	any
torsion	Member behaviour for Annex B	susceptible to torsional deformation (Table B.2)		assumption: An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).	any
$section_{class}$	Cross-section class (class 4 is outside this sheet)	1		assumption: Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).	any
$section_{type}$	Section type, which selects the buckling curve of Table 6.4 or Table 6.5	rolled I		spec: BS EN 10365 UB 457×191×98, hot-rolled	any
$moment_{mode}$	Moment diagram given as	end moments		assumption: The moments are the member end moments.	any
ltb_{curve}	Buckling curve for lateral-torsional buckling	auto (Table 6.4 / 6.5)		assumption: Curve from Table 6.5 for a rolled I with h/b > 2.	any
$curve_{y}$	Buckling curve about y-y (auto = Table 6.2 for f_y up to 420 MPa)	auto (Table 6.2)		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about y-y gives curve a	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$curve_z$	Buckling curve about z-z (auto = Table 6.2 for f_y up to 420 MPa)	auto (Table 6.2)		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about z-z gives curve b	any
use_f	Apply the modification factor f of cl. 6.3.2.3(2) to the reduction factor	yes		assumption: The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.	any
h	Overall section depth	467.2	mm	spec: BS EN 10365 UB 457×191×98	100 … 1500 mm
b	Flange width	192.8	mm	spec: BS EN 10365 UB 457×191×98	50 … 600 mm
t_f	Flange thickness, which selects the flexural buckling curves of Table 6.2	19.6	mm	spec: BS EN 10365 UB 457×191×98	5 … 60 mm
A	Cross-section area	125.265	cm²	reference: Section properties sheet SEC-01, UB 457×191×98 (A = 125.265 cm²)	10 … 1000 cm²
I_y	Second moment of area about y-y (major axis)	45728	cm⁴	reference: Section properties sheet SEC-01 (I_y = 45727.99 cm⁴)	500 … 1000000 cm⁴
I_z	Second moment of area about z-z (minor axis)	2347.02	cm⁴	reference: Section properties sheet SEC-01 (I_z = 2347.02 cm⁴)	50 … 100000 cm⁴
I_t	Torsion constant	121.287	cm⁴	reference: Section properties sheet SEC-01 (J = 121.287 cm⁴)	1 … 10000 cm⁴
I_w	Warping constant	1175540	cm⁶	reference: Section properties sheet SEC-01 (I_w = 1175538.6 cm⁶)	1000 … 1000000000 cm⁶
$W_{pl,y}$	Plastic section modulus about y-y	2232.46	cm³	reference: Section properties sheet SEC-01 (W_pl,y = 2232.46 cm³)	10 … 20000 cm³

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$W_{el,y}$	Elastic section modulus about y-y	1957.53	cm³	reference: Section properties sheet SEC-01 (W_el,y = 1957.53 cm³)	10 … 20000 cm³
$W_{pl,z}$	Plastic section modulus about z-z	378.904	cm³	reference: Section properties sheet SEC-01 (W_pl,z = 378.904 cm³)	5 … 10000 cm³
$W_{el,z}$	Elastic section modulus about z-z	243.467	cm³	reference: Section properties sheet SEC-01 (W_el,z = 243.467 cm³)	5 … 10000 cm³
L	Length between lateral restraints (fork supports)	5	m	drawing: FR-BM-12 rev B	0.5 … 50 m
f_y	Yield strength for the thickness	345	MPa	spec: BS EN 10025-2 S355, 16 mm < t ≤ 40 mm (flange 19.6 mm)	235 … 420 MPa
E	Young's modulus	210	GPa	spec: BS EN 1993-1-1 cl. 3.2.6	200 … 215 GPa
G	Shear modulus	81	GPa	spec: BS EN 1993-1-1 cl. 3.2.6: G = E/(2(1 + ν)) ≈ 81 GPa	75 … 85 GPa
N_{Ed}	Design axial force, tension positive and compression negative (a positive value is outside the range)	-400	kN	reference: S2 frame sheet FR-01, member 2, ULS combination (compression negative)	-20000 … 0 kN
$M_{z,Ed}$	Greatest design moment about z-z	6	kN·m	assumption: Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).	0 … 10000 kN·m
ψ_z	Ratio of the end moments about z-z	1		assumption: Uniform minor-axis moment (conservative).	-0.9 … 1
$M_{y,A}$	Moment at end A about y-y as an ordinate of the bending-moment diagram, with the same sign at both ends for single curvature, unlike the end actions of a stiffness analysis (optional; blank = the sheet is blocked in the end-moment mode)	340	kN·m	reference: S2 frame sheet FR-01, member 2, end A	-10000 … 10000 kN·m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$M_{y,B}$	Moment at end B about y-y, with the same convention as end A: an ordinate of the bending-moment diagram, the same sign as A for single curvature (optional; blank = the sheet is blocked in the end-moment mode)	170	kN·m	reference: S2 frame sheet FR-01, member 2, end B (same side: single curvature)	-10000 … 10000 kN·m
k	Effective length factor for end rotation about the minor axis	1		assumption: Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).	0.7 … 1
k_w	Effective length factor for warping	1		assumption: Free to warp at the restraints.	0.7 … 1
z_g	Load height above the shear centre, positive towards the compression flange (destabilising)	0	mm	assumption: End-moment loading: no transverse load, so the load height does not enter.	-500 … 500 mm
$L_{cr,y}$	Buckling length about y-y	5	m	drawing: FR-BM-12 rev B	0.5 … 50 m
$L_{cr,z}$	Buckling length about z-z	5	m	drawing: FR-BM-12 rev B	0.5 … 50 m
γ_{M1}	Partial factor for member buckling	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.1
α_{a0}	Imperfection factor for curve a0 (recommended 0.13)	0.13		spec: EN 1993-1-1 Table 6.1, recommended value	0.1 … 0.8
α_a	Imperfection factor for curve a (recommended 0.21)	0.21		spec: EN 1993-1-1 Table 6.1, recommended value	0.1 … 0.8
α_b	Imperfection factor for curve b (recommended 0.34)	0.34		spec: EN 1993-1-1 Table 6.1, recommended value	0.1 … 0.8
α_c	Imperfection factor for curve c (recommended 0.49)	0.49		spec: EN 1993-1-1 Table 6.1, recommended value	0.1 … 0.8
α_d	Imperfection factor for curve d (recommended 0.76)	0.76		spec: EN 1993-1-1 Table 6.1, recommended value	0.1 … 0.8

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
λ_{LT0}	Plateau slenderness for cl. 6.3.2.3 (recommended 0.4)	0.4		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.2 ... 0.4
β_{LT}	Slenderness correction factor for cl. 6.3.2.3 (recommended 0.75)	0.75		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.6 ... 1
C_2	Moment diagram factor C_2 (0 for end moments; enter it for a transverse load)	0		assumption: C_2 is 0 for end-moment loading.	0 ... 1

3. Assumptions

A1. The member is uniform, doubly symmetric and on fork supports, free to warp and to rotate about the minor axis. (Mono-symmetric, tapered and cantilever members and continuous restraint are not modelled.)
A2. A k or k_w below 1 models partial restraint of the end rotation or of the warping. (The factors k and k_w are entered, not derived.)
A3. The cross-section class is the class the user states, 1, 2 or 3. (Effective properties of class 4 sections are not derived.)
A4. The factor C_1 from the ψ formula applies to end-moment diagrams with k = 1. (For other diagrams the user enters C_1, C_2, k_c and C_{my} from the source in use.)
A5. No table of C_1 or C_2 values is shipped, so the user takes them from the source in use. (The factors are stored as reference records.)
A6. The imperfection factors, β and $\bar{\lambda}_{LT,0}$ default to the recommended values, which the National Annex may change. (They are a factor set the user can verify.)
A7. The 2022 edition choice changes the printed label only, and the arithmetic stays that of the 2005+A1:2014 edition. (The differences between the editions cannot be stated with certainty, and the sheet warns.)
A8. The flexural buckling curves chosen automatically are those of Table 6.2 for S 235 to S 420. (For a higher f_y or a combination without a row the user chooses the curve.)
A9. The Annex A and Annex B expressions are implemented as published. (The user checks their coefficients against the owner's copy of the standard.)
A10. $\text{ltb}_{\text{method}}$ (Method for the lateral-torsional buckling reduction factor) is taken as 6.3.2.3 (rolled and welded sections). (6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower $M_{b,Rd}$.)
A11. annex (Annex for the interaction factors) is taken as Annex B (method 2). (Annex B (method 2) chosen; Annex A can be selected for comparison.)
A12. torsion (Member behaviour for Annex B) is taken as susceptible to torsional deformation (Table B.2). (An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).)
A13. $\text{section}_{\text{class}}$ (Cross-section class (class 4 is outside this sheet)) is taken as 1. (Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).)
A14. $\text{moment}_{\text{mode}}$ (Moment diagram given as) is taken as end moments. (The moments are the member end moments.)
A15. $\text{ltb}_{\text{curve}}$ (Buckling curve for lateral-torsional buckling) is taken as auto (Table 6.4 / 6.5). (Curve from Table 6.5 for a rolled I with h/b > 2.)
A16. use_f (Apply the modification factor f of cl. 6.3.2.3(2) to the reduction factor) is taken as yes. (The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.)
A17. $M_{z,Ed}$ (Greatest design moment about z-z) is taken as 6 kN·m. (Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).)
A18. ψ_z (Ratio of the end moments about z-z) is taken as 1. (Uniform minor-axis moment (conservative).)
A19. k (Effective length factor for end rotation about the minor axis) is taken as 1. (Fork supports at the

lateral restraints (no end restraint of the minor-axis rotation).)
A20. k_w (Effective length factor for warping) is taken as 1. (Free to warp at the restraints.)
A21. z_g (Load height above the shear centre, positive towards the compression flange (destabilising)) is taken as 0 mm. (End-moment loading: no transverse load, so the load height does not enter.)
A22. C_2 (Moment diagram factor C_2 (0 for end moments; enter it for a transverse load)) is taken as 0. (C_2 is 0 for end-moment loading.)
A23. Flexural buckling curves a (y-y) and b (z-z) from Table 6.2: rolled I-section, h/b = 2.4232, t_f = 19.6 mm. (Automatic selection; choose the curves by hand to override.)

4. Working

[36] $M_{y,Ed} = \max(|M_A|, |M_B|)$ [3]
$= \max(|340|, |170|)$ kN·m
$= 340$ kN·m
Greatest design moment about y-y

[37] $\psi_y = \psi = M_{small}/M_{large}$ (same sign: single curvature) [4](B.3)
$= 170 / 340$
$= 0.5$
Ratio of the end moments, taken as ordinates of the bending-moment diagram (same sign at both ends: single curvature)

[38] $N_{abs} = abs(N_{Ed})$
= the compression force, positive
= 400 kN
Axial force magnitude

[39] $C_1 = 1.88 - 1.40\psi + 0.52\psi^2 \le 2.70$ [2](/ ENV 1993-1-1 Annex F)
$= \psi = 0.5$
$= 1.31$
C_1 for end moments (derived)

[40] $Mcr = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 I_w/I_z + (kL)^2 G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$ [2]
$= 1.31 \times \pi^2 \times 210000 \times 23470000 / (5000)^2 \times [\sqrt{((1/1)^2 \times 1175500000000/23470000 + (5000)^2 \times 81}$
$1212900 / (\pi^2 \times 210000 \times 23470000) + (0 \times 0)^2) - 0]$
$= 808.38$ kN·m
Elastic critical moment for lateral-torsional buckling

[41] $W_y = W_{pl,y}$ (class 1, 2) [5]
= plastic
$= 2232.5$ cm³
Section modulus used for bending about y

[42] $\lambda_{LT} = \sqrt{(W_y \cdot f_y/M_{cr})}$ [5]
$= \sqrt{(2232.5}$ cm³ $\times$ 345 MPa / 808.38 kN·m)
$= 0.9761$
Non-dimensional slenderness for lateral-torsional buckling

[49] α_{LT} = imperfection factor, curve c [6](6.5)
= Table 6.5, rolled, h/b = 2.4232
$= 0.49$
Imperfection factor for lateral-torsional buckling

[52] $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \cdot \bar{\lambda}_{LT}^2]$ [7]
$= 0.5 \times [1 + 0.49 \times (0.9761 - 0.4) + 0.75 \times 0.9761^2]$
$= 0.99843$
Φ_{LT}

[53] $\chi_{LT,0} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \beta \cdot \bar{\lambda}_{LT}^2)}) \le 1, \le 1/\bar{\lambda}_{LT}^2$ [7]
$= 1/(0.99843 + \sqrt{(0.99843^2 - 0.75 \times 0.9761^2)})$
$= 0.6537$
Lateral-torsional reduction factor

[55] $k_c = 1/(1.33 - 0.33\psi)$ [4](6.6)
$= \psi = 0.5$
$= 0.85837$

k_c for a linear moment diagram (derived)

[56] $f = 1 - 0.5(1 - k_c)[1 - 2(\bar{\lambda}_{LT} - 0.8)^2] \le 1$ [8]

$= 1 - 0.5 \times (1 - 0.85837) \times [1 - 2 \times (0.9761 - 0.8)^2]$

$= 0.93358$

Modification factor f

[57] $\chi_{LT,mod} = \chi_{LT}/f \le 1, \le 1/\bar{\lambda}_{LT}^2$ (eq. 6.58) [8]

$= 0.6537 / 0.93358, 1/\bar{\lambda}_{LT}^2 = 1.0496$

$= 0.70022$

Modified lateral-torsional reduction factor

[58] $\chi_{LT} = \chi_{LT,mod}$ [9]

$=$ as above

$= 0.70022$

Lateral-torsional reduction factor used

[59] $Mb_{Rd} = \chi_{LT} \cdot W_y \cdot f_y / \gamma_{M1}$ [10]

$= 0.70022 \times 2232.5\ cm^3 \times 345\ MPa / 1$

$= 539.3\ kN \cdot m$

Design buckling resistance moment

[60] $u_{LTB} = M_{y,Ed} / Mb_{Rd}$

$= 340\ kN \cdot m / 539.3\ kN \cdot m$

$= 0.63044$

Lateral-torsional buckling utilisation

[63] α_y = imperfection factor, curve a [11](6.2)

$=$ Table 6.2

$= 0.21$

Imperfection factor about y

[64] α_z = imperfection factor, curve b [11](6.2)

$=$ Table 6.2

$= 0.34$

Imperfection factor about z

[65] $N_{Rk} = A \cdot f_y$ [12]

$= 125.27\ cm^2 \times 345\ MPa$

$= 4321.6\ kN$

Characteristic axial resistance

[66] $N_{cr,y} = \pi^2 E \cdot I_y / L_{cr,y}^2$ [13]

$= \pi^2 \times 210000 \times 457280000 / 5000^2$

$= 37911\ kN$

Elastic critical force about y

[67] $N_{cr,z} = \pi^2 E \cdot I_z / L_{cr,z}^2$ [13]

$= \pi^2 \times 210000 \times 23470000 / 5000^2$

$= 1945.8\ kN$

Elastic critical force about z

[68] $\lambda_y = \sqrt{(A \cdot f_y / N_{cr,y})}$ [14]

$= \sqrt{(4321.6 / 37911)}$

$= 0.33763$

Non-dimensional slenderness about y

[69] $\lambda_z = \sqrt{(A \cdot f_y / N_{cr,z})}$ [14]

$= \sqrt{(4321.6 / 1945.8)}$

$= 1.4903$

Non-dimensional slenderness about z

[70] $\chi_y = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$ [13]

$= \alpha = 0.21, \bar{\lambda} = 0.33763$

$= 0.96853$

Reduction factor about y

[71] $\chi_z = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \leq 1$ [13]
= $\alpha = 0.34$, $\bar{\lambda} = 1.4903$
= 0.34582
Reduction factor about z
[72] $N_{b,y,Rd} = \chi_y \cdot A \cdot f_y / \gamma_{M1}$ [12]
= 0.96853 × 4321.6 / 1
= 4185.7 kN
Buckling resistance about y
[73] $N_{b,z,Rd} = \chi_z \cdot A \cdot f_y / \gamma_{M1}$ [12]
= 0.34582 × 4321.6 / 1
= 1494.5 kN
Buckling resistance about z
[74] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$
= min(4185.7 kN, 1494.5 kN)
= 1494.5 kN
Flexural buckling resistance (smaller of the two axes)
[75] $M_{y,Rk} = W_y \cdot f_y$ [15]
= 2232.5 cm³ × 345 MPa
= 770.2 kN·m
Characteristic bending resistance about y
[76] $M_{z,Rk} = W_z \cdot f_y$ ($W_{pl,z}$ class 1, 2; $W_{el,z}$ class 3) [15]
= 378.9 cm³ × 345 MPa
= 130.72 kN·m
Characteristic bending resistance about z
[77] $C_{my} = 0.6 + 0.4\psi_y \geq 0.4$ (or entered) [4](B.3)
= $\psi_y = 0.5$
= 0.8
Equivalent uniform moment factor, y
[78] $C_{mz} = 0.6 + 0.4\psi_z \geq 0.4$ (or entered) [4](B.3)
= $\psi_z = 1$
= 1
Equivalent uniform moment factor, z
[79] C_{mLT} = as C_{my} (or entered) [4](B.3)
= C_{my}
= 0.8
Equivalent uniform moment factor for lateral-torsional buckling
[80] k_{yy} = Annex B: $C_{my}(1 + (\bar{\lambda}_y - 0.2)n_y) \leq C_{my}(1 + 0.8n_y)$ (class 1, 2) [16](B.2)
= $n_y = 0.095565$
= 0.81052
Interaction factor k_{yy}
[81] k_{yz} = Annex B: $0.6k_{zz}$ (class 1, 2), k_{zz} (class 3) [16](B.2)
= as above
= 0.82482
Interaction factor k_{yz}
[82] k_{zy} = Annex B: Table B.1 $0.6k_{yy}$ / $0.8k_{yy}$; Table B.2 from $\bar{\lambda}_z$, n_z and C_{mLT} [16](B.2)
= $n_z = 0.26764$
= 0.95134
Interaction factor k_{zy}
[83] k_{zz} = Annex B: $C_{mz}(1 + (2\bar{\lambda}_z - 0.6)n_z) \leq C_{mz}(1 + 1.4n_z)$ (class 1, 2) [16](B.2)
= as above
= 1.3747
Interaction factor k_{zz}

[84] $u_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$ [17]
$= 400\ kN / (0.96853 \times 4321.6\ kN / 1)$
$= 0.095565$
Axial term about y
[85] $u_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$ [17]
$= 400\ kN / (0.34582 \times 4321.6\ kN / 1)$
$= 0.26764$
Axial term about z
[86] $u_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$ [17]
$= 340\ kN{\cdot}m / (0.70022 \times 770.2\ kN{\cdot}m / 1)$
$= 0.63044$
Major-axis bending term
[87] $u_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$ [17]
$= 6\ kN{\cdot}m / (130.72\ kN{\cdot}m / 1)$
$= 0.045899$
Minor-axis bending term
[88] $u_{61} = N_{Ed}/(\chi_y\ N_{Rk}/\gamma_{M1}) + k_{yy} \cdot M_{y,Ed}/(\chi_{LT}\ M_{y,Rk}/\gamma_{M1}) + k_{yz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$ [17]
$= 0.095565 + 0.81052 \times 0.63044 + 0.82482 \times 0.045899$
$= 0.64441$
Interaction, equation 6.61
[89] $u_{62} = N_{Ed}/(\chi_z\ N_{Rk}/\gamma_{M1}) + k_{zy} \cdot M_{y,Ed}/(\chi_{LT}\ M_{y,Rk}/\gamma_{M1}) + k_{zz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$ [17]
$= 0.26764 + 0.95134 \times 0.63044 + 1.3747 \times 0.045899$
$= 0.9305$
Interaction, equation 6.62
[90] $u_{interaction} = \max(u_{61}, u_{62})$ [17]
$= \max(0.64441, 0.9305)$
$= 0.9305$
Greatest interaction ratio
[91] $u_{buckling} = N_{Ed}/N_{b,Rd}$ [18]
$= 400\ kN / 1494.5\ kN$
$= 0.26764$
Flexural buckling utilisation
[92] $u_{member} = \max(\max(u_{interaction}, u_{LTB}), u_{buckling})$
$= \max(\max(0.9305, 0.63044), 0.26764)$
$= 0.9305$
Governing utilisation of the member
[93] $u_{limit} = limit$
$= the\ utilisation\ limit$
$= 1$
Utilisation limit

5. Results

SYMBOL	VALUE	DESCRIPTION
Mb_{Rd}	539.30 kN·m	$M_{b,Rd}$
Mcr	808.38 kN·m	M_{cr}
χ_{LT}	0.70022	χ_{LT}
λ_{LT}	0.9761	$\bar{\lambda}_{LT}$
$N_{b,Rd}$	1494.5 kN	$N_{b,Rd}$
u_{LTB}	0.63	LTB utilisation
$u_{interaction}$	0.93	Interaction utilisation

SYMBOL	VALUE	DESCRIPTION
u_{member}	0.93	Member utilisation
$edition_{used}$	"EN 1993-1-1:2005+A1:2014"	EN 1993-1-1 edition
ψ_y	0.5	ψ of the M_y diagram
$M_{y,Ed}$	340 kN·m	Greatest design moment about y-y
C_1	1.31	C_1
N_{Rk}	4321.6 kN	Characteristic axial resistance
$N_{cr,y}$	37911 kN	$N_{cr,y}$
$N_{cr,z}$	1945.8 kN	$N_{cr,z}$
λ_y	0.33763	$\bar{\lambda}_y$
λ_z	1.4903	$\bar{\lambda}_z$
χ_y	0.96853	χ_y
χ_z	0.34582	χ_z
$N_{b,y,Rd}$	4185.7 kN	$N_{b,y,Rd}$
$N_{b,z,Rd}$	1494.5 kN	$N_{b,z,Rd}$
k_{yy}	0.81052	Interaction factor for major-axis bending in equation 6.61
k_{yz}	0.82482	Interaction factor for minor-axis bending in equation 6.61
k_{zy}	0.95134	Interaction factor for major-axis bending in equation 6.62
k_{zz}	1.3747	Interaction factor for minor-axis bending in equation 6.62
α_{LT}	0.49	α_{LT}
α_y	0.21	α_y
α_z	0.34	α_z
Φ_{LT}	0.99843	Φ_{LT}
$\chi_{LT,0}$	0.6537	χ_{LT} before the f modification
k_c	0.85837	k_c
f	0.93358	f
W_y	2232.5 cm³	Section modulus for bending about y-y
$M_{y,Rk}$	770.20 kN·m	$M_{y,Rk}$
$M_{z,Rk}$	130.72 kN·m	$M_{z,Rk}$
C_{my}	0.8	Equivalent uniform moment factor about y-y
C_{mz}	1	Equivalent uniform moment factor about z-z
C_{mLT}	0.8	Equivalent uniform moment factor for lateral-torsional buckling
governing	"interaction 6.62 (0.931)"	Governing check

SYMBOL	VALUE	DESCRIPTION
$u_{buckling}$	0.27	Buckling utilisation
u_{61}	0.64	Equation 6.61
u_{62}	0.93	Equation 6.62 (governing)
$u_{N,y}$	0.10	Axial term, y
$u_{N,z}$	0.27	Axial term, z
$u_{M,y}$	0.63	Bending term, y
$u_{M,z}$	0.05	Bending term, z

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Interaction equation 6.62 against its limit	0.9305	1	0.93	[19](6.3.3(4))
PASS	Interaction equation 6.61 against its limit	0.64441	1	0.64	[19](6.3.3(4))
PASS	Bending moment against lateral-torsional buckling resistance	340 kN·m	539.30 kN·m	0.63	[19](6.3.2.1)
PASS	Axial force against flexural buckling resistance	400 kN	1494.5 kN	0.27	[19](6.3.1.1)

7. Applicability

WARNING records: tabulated values not taken from a reference record are E, G, f_y (Material properties); γ_{M1} (EN 1993-1-1 National Annex factors); α_{a0}, α_a, α_b, α_c, α_d, λ_{LT0}, β_{LT} (Imperfection factors and 6.3.2.3 parameters); C_2 (C_1, C_2 and k_c for the moment diagram). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

```
Template struct.ltb-interaction@1.3.2 Content hash 24861dbf6d477993 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. NCCI SN003.
3. 6.3.2.
4. Table.
5. 6.3.2.2(1).
6. 6.3.2.3(1), Table.
7. 6.3.2.3(1).
8. 6.3.2.3(2).
9. 6.3.2.3.
10. 6.3.2.1(3).
11. 6.3.1.2, Table.

12. 6.3.1.1(3).
13. 6.3.1.2.
14. 6.3.1.3.
15. 6.3.3.
16. Annex B Table.
17. 6.3.3(4).
18. 6.3.1.1.
19. cl.